B. Appendix Inspection

The following tables list inspection plans for the doors. The overview of the service intervals are provided in section B 1. The numbers are used in the inspection plan 0.

Information for RR300 Clean doors for GMP-relevant aspects are provided in section B 2.

B 1. Service intervals

1	half-yearly or 50.000 cycles (* only for RR300 Plus F+R)
2	annually or 50.000 cycles
3	annually or 100.000 cycles
4	annually
5	clean if necessary
6	every 4 years
7	150.000 cycles
8	250.000 cycles
9	500.000 cycles
10	150.000 cycles; RR300 Plus F+R: 100.000 cycles
11	with MCC: 150.000 cycles; with ACS50: 50.000 cycles
12	RR450Ex / 600Ex: 200.000 cycles
13	450.000 cycles; RR450Ex/600Ex 200.000 cycles
14	200.000 cycles; RR600 Crash: 100.000 cycles
15	DH < 3m: 500.000 cycles; DH ≥ 3m: 250.000 cycles
16	DH < 4m: 300.000 cycles; DH ≥ 4m: 200.000 cycles
**	Exchange of bearings only by drive manufacturer or an approved workshop for the maintenance of the explosion-proofness. (downtime at the time of external execution of the exchange, which can be expected, approx. 8-10 working days)
***	Regular conditions: central European climate, door mounted interior or exterior with covers, no aggressive ambient conditions

≤150.000 cycles: ±10.000 cycles; >150.000 cycles: ±20.000 cycles

B 2. GMP-relevant inspection (only RR300 Clean)

Could be done also by the customer.

Mechanical components	**proposal**
Check air leakage: Curtain should permit only small air leakage around gaps	3 months or 25000 cycles
Motor sealing: Motor cover should fit firm and tight, cable glands must be orderly	3 months or 25000 cycles

B 3. Inspection plan

Inspection	RR300,RR200	RR300 Plus	RR300 Clean	RR300 Food	RR355	RR392	RR450	RR500	RR600	RR3000,R,Vision	RR3000 XXL	RR3000 ISO
Installation fixings												
Tighten screws	1	1	2	1	1	1	1	1	1	3	2	3
Check welding seams	1	1	2	1	1	1	1	1	1	3	2	3
Drive												
Check the fixing of the torque arm and buffer	1	1	2	1	1	1	1	1	1	3	2	3
Check brake condition and function of break release lever	1	1	2	1	1	1	1	1	1	3	2	3
Check operation smoothness and function of the emergency manual lever, maybe lubricate		1		1	1	1	1		1	3	2	3
Check the chain, adjust and possibly retension and lubricate					1	1	1	1		3	2	2
RR450Ex / 600Ex: Change bearings in motor**:							6					
Drive shaft												
Check screws to drive shafts and flange bearings	1	1	2	1	1	1		1	1	3	2	3
Lubricate pedestal bearing and retighten it's fixation							1					
Check the drive shaft in the aluminium shaft for secure fit	1	1	2	1	1	1	1					
Check tight position of belt and rope discs		1		1	1	1	1	1				
Check the condition of the cable drums									1			
Check retainer of spiral disk on the shaft										3	2	3
Chain drive: check fastening of locking element, tighten the screws with 18 Nm										3	2	2
Check condition of sound isolation profiles at the spiral disks										3	2	2
Check flat belts for wear and damage; check connection to spiral disks										3	2	2
Door curtain												
Check curtain for cracks and traces of scrapes or shavings in the guides	1	1	2	1	1	1	1	1	1	3	2	2
Check fixation of door leaf at the top roll and at the bottom profile	1	1	2	1	1	1	1	1	1			
Check tight position and wear & tear of the wind tabs	1	1	2	1	1	1	1	1	1			
Belt: condition, lubrication, wear; if necessary replace teeth	1							1				
Check exact position of slat sealing										3	2	3
Check fixing of slats on pulling device and endcaps										3	2	2

Inspection	RR300,RR200	RR300 Plus	RR300 Clean	RR300 Food	RR355	RR392	RR450	RR500	RR600	RR3000,R,Vision	RR3000 XXL	RR3000 ISO
Check slats around adapters for damage (especially 1-5 from top)										3	2	2
Check solid position of slat adapter										3	2	3
Bottom profile												
Check tight fit and condition of profile, guiding pieces and the gliding pieces	1	1	2	1	1	1	1		1			
Check condition of the safety edge and the cable	1	1	2	1	1	1	1	1	1			
Check condition of the coily cable	1	1			1	1			1			
Check the condition and function of the crash system	1	1		1	1	1	1	1	1			
Check condition and function of wireless safety edge; change battery if necessary	7	7						7				
side frame												
Check fixing and condition of the profiles with attachments	1	1	2	1	1	1	1	1	1	3	2	3
Check cable laying and if applicable cable chain	1	1	2	1	1	1	1	1	1	3	2	3
Remove dirt from the photocell optics	5	5	5	5	5	5	5	5	5	5	5	5
Check condition, fixing and lengthwise extension of the ropes and belts		1		1	1	1	1	1	1	3	2	2
Check the condition and function of the counterweight system		1		1	1	1	1	1	1	3	2	2
Check condition and installation of the springs and the spring mounting, if applicable check for easy movement of the axial bearing		1			1		1		1			
Doors with ripped belt / rope or broken springs are to be immediately taken out of service		Yes		Yes	Yes	Yes	Yes	Yes	Yes			
Check leaf tensioning system		1			1	1	1					
Check for wear & tear (s) on guiding rails, especially in the area of funnel (s< 4mm)								1		3	2	3
Renew belt/s		8					8	8		15	16	16
Renew spring/s		10			9		13		9	15	16	16
Renew ropes		10		8	7		8					
Renew ropes and pulleys for counterbalance				9		11	12		14			
Renew ropes or rubber cable for curtain tension						8	8			9		
Renew flat belt***										9	9	9
Grease plastic guidings								2				

Inspection	RR300,RR200	RR300 Plus	RR300 Clean	RR300 Food	RR355	RR392	RR450	RR500	RR600	RR3000,R,Vision	RR3000 XXL	RR3000 ISO
Control box and additional attachments (e.g. impulse activator)												
Check for full number/completeness of circuit diagrams	4	4	4	4	4	4	4	4	4	4	4	4
Check main switch/CEE-plug and control box lock	4	4	4	4	4	4	4	4	4	4	4	4
Check the fixing	4	4	4	4	4	4	4	4	4	4	4	4
Electrical components												
Visual inspection for mechanical damage	1	1	2	1	1	1	1	1	1	3	2	2
Check potential balance	4	4	4	4	4	4	4	4	4	4	4	4
Safety equipment and impulse activators												
Measure self-opening height after switching off the door and document in the inspection logbook (minimum self-opening height: 2m)		1*										
Check self-test function by switching the key switch or switching the door On/Off		1*										
Function												
Check function of safety edge, photocell, additional safeties and actuators	1	1	1	1	1	1	1	1	1	1	1	1
Check function and condition of the torque arm during operation	1	1	2	1	1	1	1	1	1	3	2	2
Assess possibility of emergency opening	1	1	2	1	1	1	1	1	1	3	2	2
Check opening and closing position of the bottom profile	1	1	2	1	1	1	1	1	1	3	2	2
Check winding performance of door curtain and ropes/belts	1	1	2	1	1	1	1	1	1	3	2	2
Check function of belt and spring break switch		1*										
Check function of magnetic switch at 2m door height		1*										
Check radar motion detectors for mechanical damage and that they function properly (required detection range: whole door width, up to 1.5m in front of the door)		1*										
Check main closing edge according to EN 13241-1 respectively local standards	4	4	4	4	4	4	4	4	4	4	4	4

C. Appendix Cleaning reference RR300 Clean

The door could be cleaned with commercially available cleaning agents and disinfectant for clean room applications.

CAUTION



Risk of material damage

Improper cleaning of the door may cause corrosion and discoloration of the surface.

⇨ Do not use chlorous cleaning agents and disinfectant.

CAUTION



Risk of injury

Door movements during cleaning could cause injuries.

⇨ Set the main switch (power switch) to “Off” or “0” or disconnect the CEE plug, and secure the switch or plug against accidental re-activation or reconnection.



The maintenance and inspection work must only be performed by manufacturer trained/ authorized personnel.

The basic cleaning may only be performed by trained personnel. We recommend this to perform as part of the inspection.

Always follow the safety instructions mentioned in the user manual.

Maintenance cleaning
Clean outer surfaces with cleaning agent and disinfectant
Basic cleaning (during inspection)
Open side frames and clean side frame guides and trailing chain for cable
Clean lintel profile
Open top roll and motor cover and clean this area
Mount everything in reverse order

Periods need to be adjusted according to the cleaning concept and production relevance.